

Module 2 Project: Design and Implementation of a Network Solution.

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Executive Summary:

For this project, I was hired by TechLink Solutions, which is a mid-sized company, that wants to build a scalable, secure, and more efficient office network. To do this I was tasked with Segmenting the network using subnets, enabling seamless communication between departments, preparing the network for any future expansion, and reducing waste.

It was required that we:

- Use IP range: **192.168.0.0/24**
- Create subnets for PCs, printers, servers, and router-to-router communication
- Use **DHCP** for dynamic IPs and **static IPs** for infrastructure devices
- Implement **RIPv2** for dynamic routing
- Verify full connectivity across the network

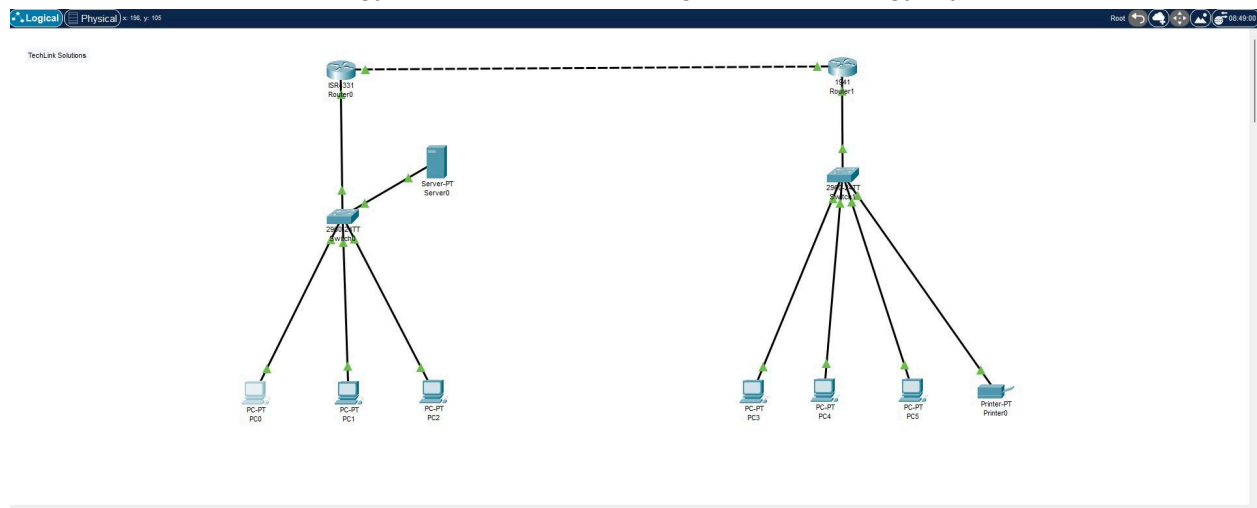
To accomplish these goals I used Cisco Packet Tracer to simulate the company, then added all the necessary components, such as switches, DHCP servers, routers, and PCs. I then assigned the static and dynamic IPs, did a RIPv2 setup, and streamlined the configuration in a manner that is efficient and supports future expansion.

In the Cisco Packet Tracer, I dragged out 2 Routers (Router0 and Router1), 2 Switches, 6 PCs (PC0-PC5), 1 Printer, and 1 Server. In this project, cables were used to connected the Routers to each other, and to their switches. Then cables were used to connect the server, printers and PCs to the two switches. I configured 3 PCs per Router, a DHCP server was the primary distributor of the IPs Addresses of the PCs. To do that I had to configure each router so that both had their own address and know what address to dish out to the other devices in its respective network. I also used RIPv2 to make sure that the devices from each router could be in communication. To ensure that the RIPv2 was successful I then made sure that each pc and both servers could talk with one another using the ping method.

Network Topology Diagram

Each Router (Router0 and Router1) was placed at the top with one cable connecting Router0 to Router1. Each router was connected to its respective switch (switch0 and Switch1). Server0 was connected to Switch0 and was configured to a DHCP server which assigns a IP addresses to the devices in the networks. Then PC0, PC1, and PC2 were connected to Switch0 using automatically choose connection type option (which was the Cooper straight-through cables). PC3, PC4, and PC5 were connected physically to Switch1 using the same method. The PCs were connected/setup using DHCP which game each an assigned IP address with the allowed range of IPs that were set by the DHCP server and routers. The printer which was connected by cooper straight-through cable was given a static IP address so that it would always have the same one. I build this topology with the goal of an efficient setup as well as leaving room for company growth in the future.

Here is the Network Topology chart as well as a image of the Topology layout:



Subnetting Scheme

Subnet ID	Subnet Mask	Require d Hosts	Usable IP Range	Network Address	Broadcast Address	Devic es in Subne t
192.168.0.0	255.255.255.1 92 (/26)	~62	192.168.0.1 — 192.168.0.6 2	192.168.0.0	192.168.0.6 3	Router 0 (LAN), PC0, PC1, PC2, Printer

						Server
192.168.0.6 4	255.255.255.1 92 (/26)	~62	192.168.0.6 5 – 192.168.0.1 26	192.168.0.6 4	192.168.0.1 27	Router 1 (LAN), PC3, PC4, PC5
192.168.0.1 28	255.255.255.2 52 (/30)	2	192.168.0.1 29 – 192.168.0.1 30	192.168.0.1 28	192.168.0.1 31	Router 0 ↔ Router 1 (WAN link)

Device Configurations

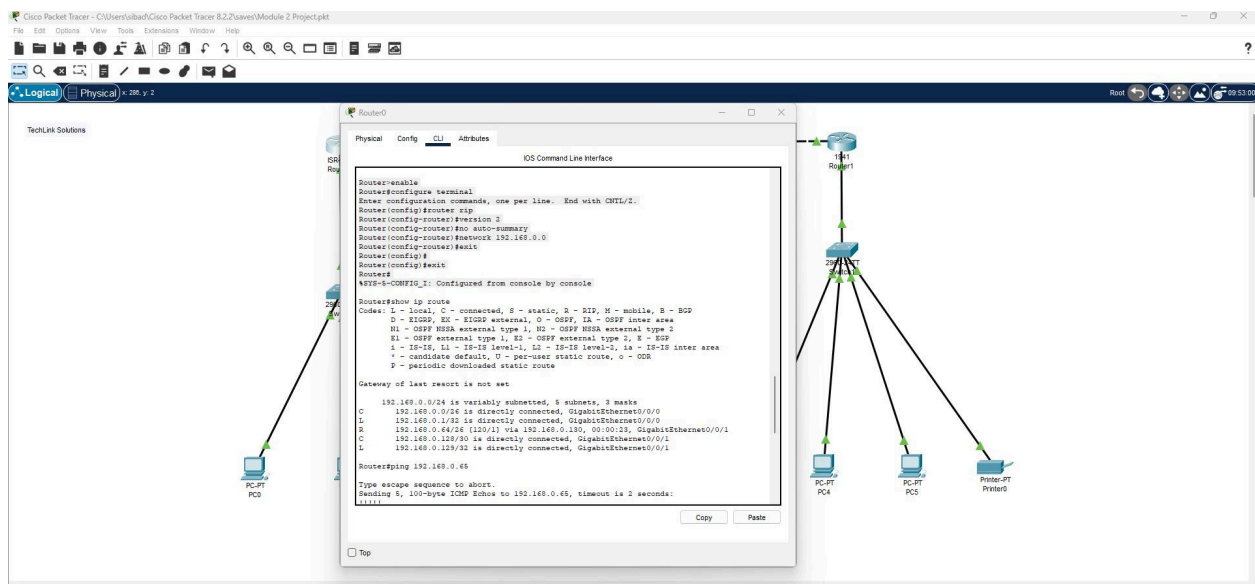
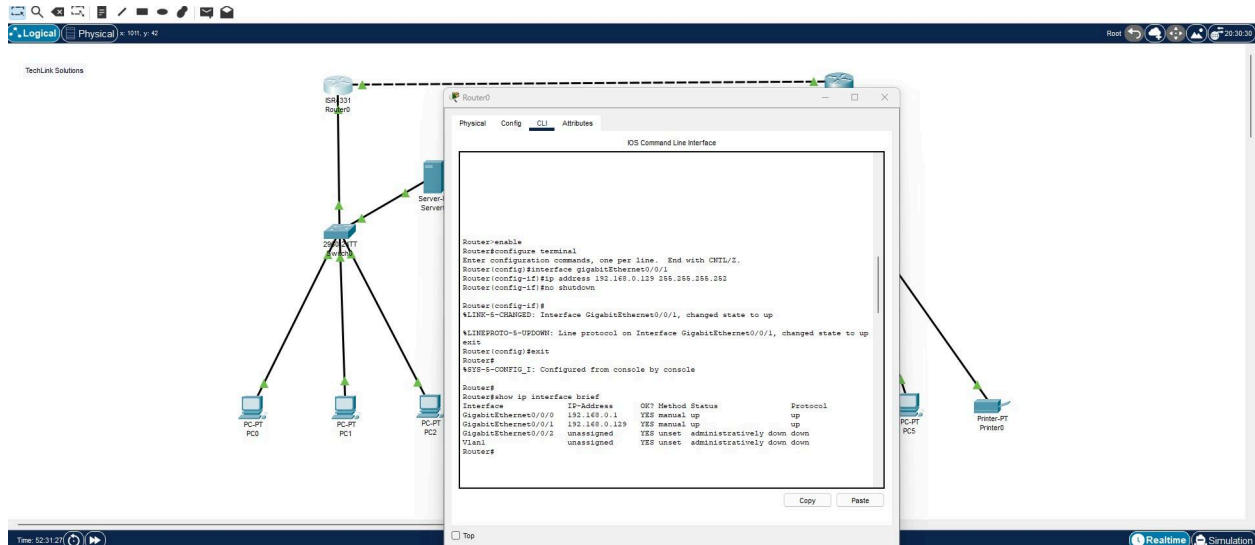
- **Router Config:**
 - RIPv2 network statements
 - IP address assignment

Let's talk about how I Configured Router0. I started out by going to CLI tab of Router0 and then assigned IPs to its interfaces:

Gig0/0/0 = 192.168.0.1. 255.255.255.192 (for LAN with PC0 - PC2)

Gig0/0/1 = 192.168.0.129. 255.255.255.252 (For WAN link to Router1)

I also made sure to input No shutdown on both interfaces.



Here is the Commands/RIPv2 Network Statements:

RIPv2 process and information:

Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#router rip

Router(config-router)#version 2

Router(config-router)#no auto-summary

Router(config-router)#network 192.168.0.0

Router(config-router)#exit

Router(config)#

Router(config)#exit

Router#

%SYS-5-CONFIG_I: Configured from console by console

On both routers:

router rip

version 2

no auto-summary

network 192.168.0.0

- **Switch Config:**

- VLANs (optional)
- Security settings

For this assignment we used basic switches so no extra VLAN or security settings were added. Port security could have been added so that only certain MAC addresses are allowed.

- **DHCP Config:**

- Dynamic pool for clients
- Static reservations

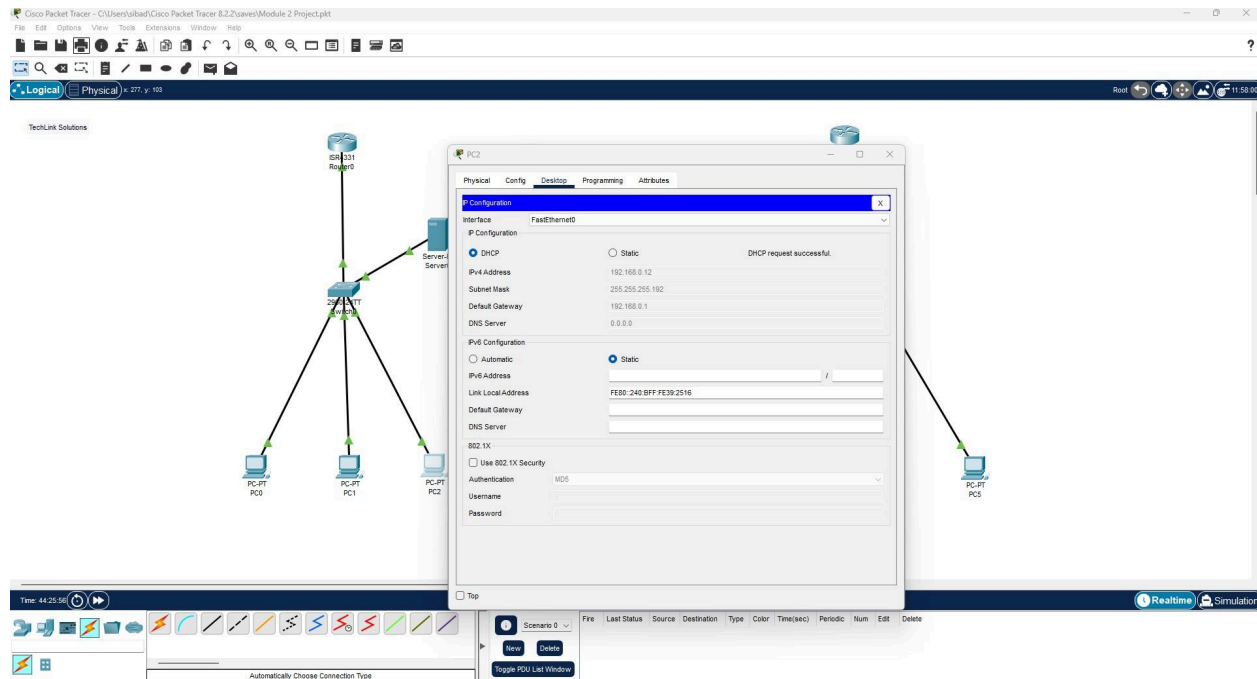
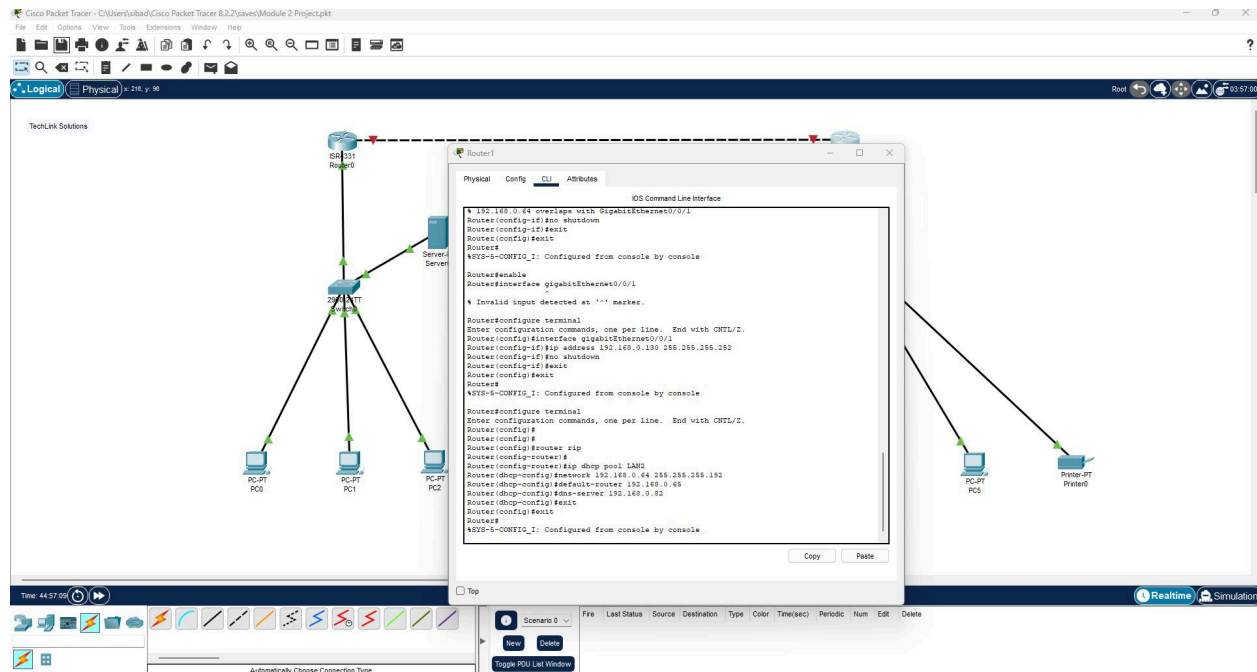
As far as creating a DHCP pool for LAN1 here is what I entered into Router0:

ip dhcp pool LAN1

network 192.168.0.0 255.255.255.192

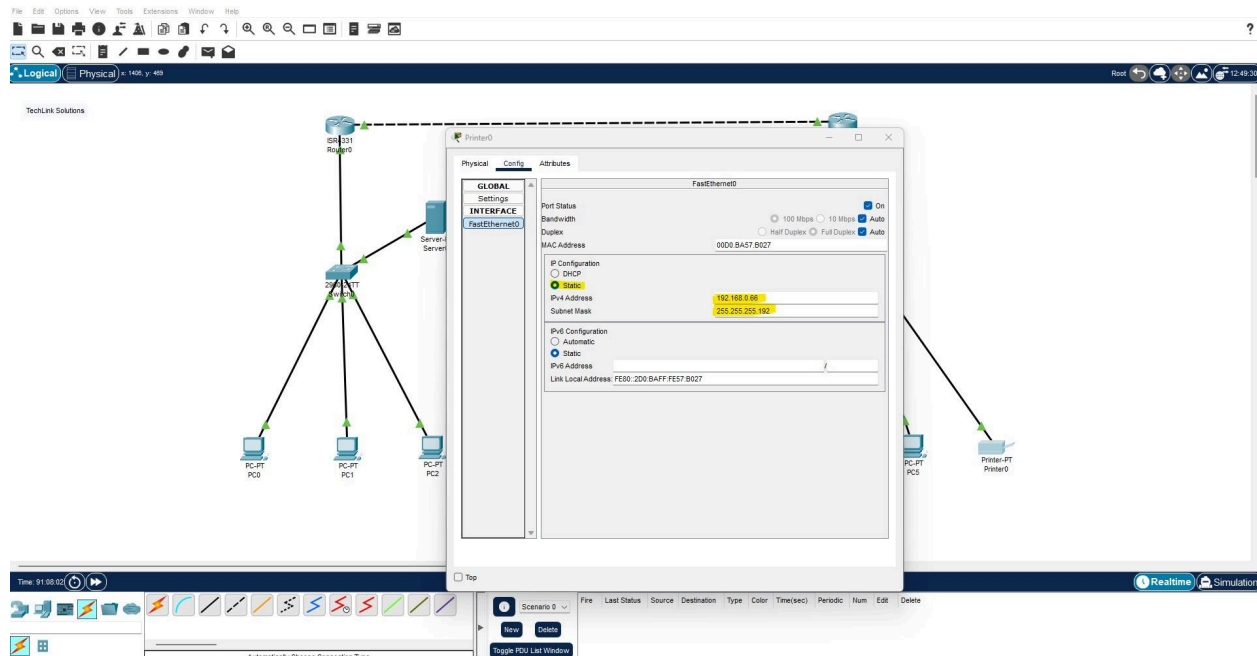
default-router 192.168.0.1

dns-server 192.168.0.82



- **Static IP Assignments:**
 - Servers, printers, and router interfaces

For this we used static IPS for the printer and the server. We did this so that the IP addresses for these items would always stay the same as opposed to having them change over time which can lead to inefficiencies.



Setting UP the Server0:

So I set the Server for Router0 as a DHCP Server. I did this by going using the auto connect feature to connect the Server to the Router. I then went into the Server and configured the DHCP settings with the settings below:

Default Gateway - 192.168.0.1,
Start IP Address - 192.168.0.10,
Subnet Mask - 255.255.255.192
Max Users as 50

Since a DHCP automatically assigns IP addresses, subnet masks, default gateways, and DNS servers to PCs, this will save time from having to manually assigning IPs to each PC, reduces configuration errors, and is ideal for large networks where devices may come and go (like employee laptops or desktops).

Cisco Packet Tracer - C:\Users\labad\Cisco Packet Tracer 8.2.2\user\Module 2 Project.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical 11:24:00

TechLink Solutions

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